

Video 1: MECH5200 Numerical Methods for PDEs : *Introduction to MATLAB and Vectors*

Valentin Boutrouche

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What is the goal of this video?

- What is a vector?
- How to construct vectors in MATLAB command prompt.
- MATLAB & vectors: Accessing vector entries via an index.
- Vector magnitude, L_2 -norm
- Vector addition, scaling operations.
- M-Independent vectors will span the M-Dimensional space
- The (vector) dot product.

Introduction to Vectors

- A vector with N entries describes a *direction* and *magnitude* in an N -dimensional space:

$$V_{col} = \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_n \end{bmatrix} \quad (1)$$

- : Or we can write this as a row-vector:

$$V_{row} = [V_1, V_2, \dots, V_n] \quad (2)$$

- *Example:* Lets declare a few $N = 2$, therefore 2-D vectors:

$$SampleVector1 = [2, 3] \quad SampleVector2 = [1, 6]$$

$$SampleVector3 = [1, 0] \quad SampleVector4 = [0, 1]$$

- In this example, the vectors each have an x -component and a y -component ($SampleVectorN = [x_n, y_n]$).
- Let's plot the vectors in MATLAB using the command:
`plot([0, SampleVector1(1)], [0, SampleVector1(2)], 'ro-')`.

Vectors: Accessing Vector Entries in MATLAB

- How do we access and manipulate individual values in a vector when using MATLAB?
- Example: let's declare a vector:

$$\text{vect6} = [1, 2, 3, 4, 5, 6]; \quad (3)$$

- **Try this:** $\text{vect6}(1) = 4;$
- **Try this:** Cycle over values and augment the entry by the value index, e.g.: $\text{vect6}(3) = \text{vect6}(3) + 3$
- **Try this:** Cycle over the values and subtract the value of the previous vector entry from the current one.

Vectors: Length and Magnitude of a Vector

- **length(V)** of the vector in MATLAB – determines the # of elements/entries in the vector.
- **Magnitude:** The **Euclidean length** or magnitude of a vector in MATLAB:

$$||\text{SampleVector1}||_2 = \sqrt{\text{SampleVector1}_x^2 + \text{SampleVector1}_y^2} \quad (4)$$

- Or, more generally:

$$||\text{SampleVector1}||_2 = \sqrt{\sum \text{SampleVector1}_i^2} \quad (5)$$

- MATLAB: We can quickly calculate the magnitude of a vector using MATLAB ($L_1 = \text{norm}(\text{SampleVector1}, 2)$).
- We'll talk later about **norms**, but the L_2 norm of a vector is the Euclidean length or Magnitude.

Vectors: Addition and Subtraction of Vectors

- Let's start manipulating the vectors.
- By *scaling* and *adding* 2 independent 2D row-vectors, $\rightarrow$ we get a 3rd vector in the space:

$$\begin{aligned} \text{NewVector} &= 3.0 * (\text{SampleVector1}) - 0.5 * (\text{SampleVector2}) \\ &= 3.0 * [2, 3] - 0.5 * [1, 6] \\ &= [6, 9] - [0.5, 3] \\ &= [5.5, 6] \end{aligned}$$

- This could also be written as an expression using **column vectors**:

$$\text{NewVector} = \alpha \cdot \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \beta \cdot \begin{bmatrix} 1 \\ 6 \end{bmatrix} = 3.0 \cdot \begin{bmatrix} 2 \\ 3 \end{bmatrix} - 0.5 \cdot \begin{bmatrix} 1 \\ 6 \end{bmatrix} = \begin{bmatrix} 5.5 \\ 6 \end{bmatrix}$$

- Let's look at a visual representation of this vector scaling and addition.

Vectors: Spanning 2D Space with 2 Independent 2D Vectors

- **Visually interpreting:** A pair of *linearly independent* vectors can be scaled and added to produce any vector in the 2D space/plane defined by those vectors.

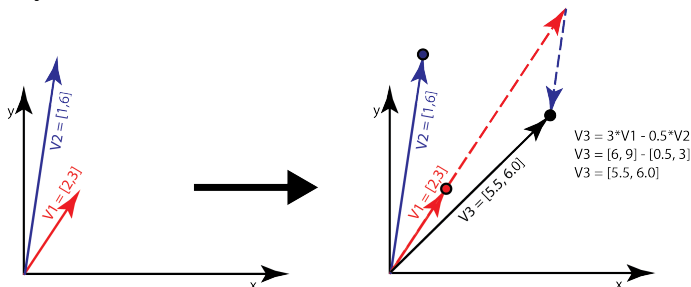


Figure: Adding two linearly independent vectors produces a 3rd vector in the 2D space.

- If the M-vectors are not independent, they won't span the M-D space.

Vectors: Spanning 2D Space with 2 Independent 2D Vectors

- **MATLAB:** Can we extend this example to “demonstrate” (numerically using MATLAB) that the scaling and addition of 2 linearly independent vectors will span the whole 2D space defined by the 2-vectors?
- Spoiler alert: Yes, we can!
- Can we show that when we have 3 vectors in 2D space, at least one of these vectors will be a linear combination of the other two?
- Spoiler alert: Yes, we can!

Vectors: 2-independent vectors: N -dimensional space

Before writing any code, let's organize our thoughts in "Pseudo Code":

Pseudo Code: A simplified version of the computer code that can assist with code design:

- Define the two linearly independent vectors:
 - **SampleVector1**
 - **SampleVector2**
- Loop k -times and introduce a random scaling **randScale1**, **randScale2** that will scale **SampleVector1** and **SampleVector2** respectively.
 - Scale the vectors:
 - **TempVec1** $\leftarrow$ **randScale1*****SampleVector1**
 - **TempVec2** $\leftarrow$ **randScale2*****SampleVector2**
 - Add together the two vectors:
 - **SumScaleVec** $\leftarrow$ **TempVec1** + **TempVec2**
- Plot the result on an x, y plot:
plot([0, SumScaleVec(1)], [0,SumScaleVec(2)],'-ok'); **hold on;**

Multiplication of Vectors: Dot Product/Inner Product

Summary of implementation results/observations:

- The original 2 x vectors when randomly scaled and added together produce a new vector in the 2-D plane defined by the original vectors.
- If all possible scaling values were considered, the two original vectors would produce all possible vectors in the 2D space defined by the original vectors.
- When the original 2 x vectors are *linearly dependent* (pointed in the same direction) the result is that the scaled combination of the vectors can only represent a 1D space (defined by the original vectors).
- This result will be used when we consider the concept of column space/column vector independence in matrices.

Multiplication of Vectors: Dot Product/Inner Product

- How do we multiply two vectors, and what does it mean when we do?
- A common vector multiplication is a **dot product**
- Consider two vectors (a , and b), the *dot – product* or *inner – product*:

$$a \cdot b = a_1 * b_1 + a_2 * b_2 + \dots + a_n * b_n \quad (6)$$

- The result of a dot product is a scalar value (single value).

Multiplication of Vectors: Dot Product/Inner Product

- **Let's try this pout in MATLAB.**

- For example, $a \cdot b$ can be written as:

```
a = [1, 2, 3]; b = [3, 2, 1];  
dot_prod = 0;  
for i = 1:length(a)  
    dot_prod = dot_prod + a(i)*b(i);  
end
```

- **or:**

```
dot_prod = a*b';  
% row vector a multiplied by a column vector, b'  
% this is using vectorization MATLAB,
```

- **or:**

```
dot_prod = dot(a,b);  
% this is using internal compiled code
```

Multiplication of Vectors: Dot Product or Inner Product

What is the meaning of these inner products of vectors?

Multiplication of Vectors: Dot Product or Inner Product

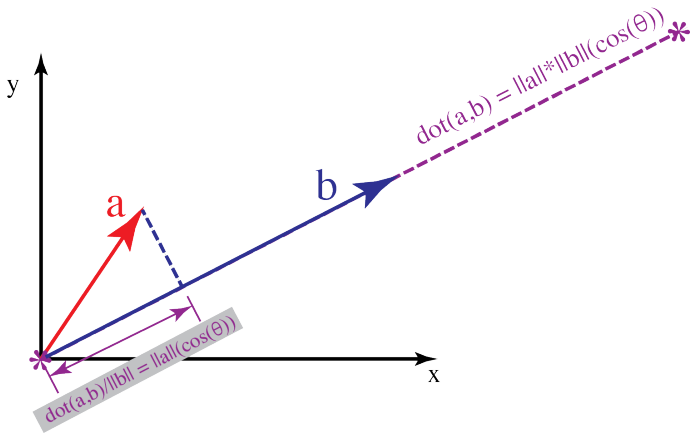


Figure: Representation # 1 for the dot product as a projection operator. $\vec{a}$ is projected onto $\vec{b}$ during the multiplication (or vice versa).

Multiplication of Vectors: Dot Product or Inner Product

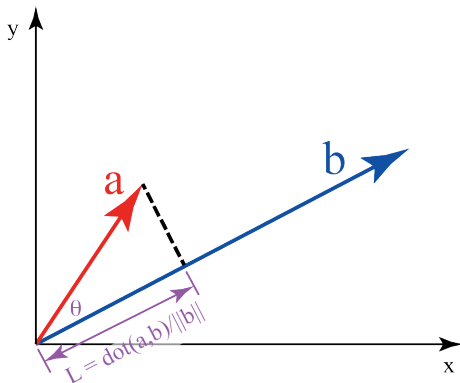


Figure: Representation # 2 the dot product when normalized by length of one vector = projection. $\vec{a}$ is projected onto $\vec{b}$ or vice versa during this operation.

Summary: What have we learned?

- A vector is a list of numbers in an N -Dimensional space.
 - **Magnitude**: Vectors have a magnitude or length
 - **Direction**: Vectors have a direction or angle
- Accessing vector entries is easy in MATLAB. One can use and index a vector variable and cycle over the elements of a vector easily (e.g.: `for(i = 1:length(A))`).
- $M \times$ linearly independent vectors can be added/subtracted/scaled to span the M -Dimensional space
- Vectors can be multiplied together in a **dot product** operation.
 - This is a **projection** operation (one vector is projected onto the other during the multiplication operation to produce a scalar multiplication as a result).